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SUCTION UNIT WITH COOLING USING FRESH AIR

Abstract

A suction unit for a handheld vacuum appliance contains a blower configured to generate a main suction air stream from the suction mouth via a separating unit to the blower of the suction unit. The suction unit has a handle with a cavity which is formed by a housing wall and in which an electronics component is arranged. The cavity has an inlet opening in the housing wall. The suction unit has a bypass channel which has an inlet facing toward the cavity and an outlet facing toward the blower. The blower is configured to generate a bypass air stream from the inlet opening, through the cavity and through the bypass channel to the blower. The cavity is covered by a shielding element which is configured to shield the cavity against moisture from the main suction air stream and/or from exhaust air from the blower.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority, under 35 U.S.C. § 119, of German Patent Application DE 10 2024 202 005.2, filed Mar. 4, 2024; the prior application is herewith incorporated by reference in its entirety.

FIELD AND BACKGROUND OF THE INVENTION

[0002] The invention relates to a suction unit for a vacuum appliance, in particular for a wireless and/or handheld vacuum mop.

[0003] A vacuum appliance, in particular a handheld vacuum cleaner, typically contains a suction unit that a user can carry and control by hand. The suction unit has a blower that is operated using electrical energy from an electrical energy store of the suction unit. The blower is configured to generate a suction air stream in order to suction contaminants through a suction mouth of the suction unit into a collecting container of the suction unit. The suction mouth of the suction unit is typically configured as a coupling by means of which in each case one accessory part from a set of different accessory parts can be connected to the suction unit. An example of an accessory part is a wet cleaning nozzle, which makes it possible to simultaneously mop and vacuum a surface that is to be cleaned.

[0004] If the vacuum appliance is used as a combined vacuum mop, the suction air stream may contain moisture, and the suction air stream can therefore no longer be directly used to cool the electronics of the suction unit.

[0005] Published, non-prosecuted German patent application DE 17 03 970 A describes a vacuum cleaner having an inlet hole for ambient air. Published, European patent application EP 3 763 265 A1 describes a portable vacuum apparatus.

[0006] Published, non-prosecuted German patent application DE 10 2012 207 348 A1 describes a vacuum cleaner having a secondary air channel. Published, non-prosecuted German patent application DE 10 2014 200 663 A1 (corresponding to U.S. Pat. No. 9,370,285) describes a battery-powered handheld vacuum cleaner. German utility model DE 20 2012 101 781 U1 describes a vacuum apparatus for hard surfaces.

SUMMARY OF THE INVENTION

[0007] The present document is concerned with the technical problem of achieving efficient and reliable cooling of the electronics of the suction unit of a vacuum appliance, in particular of a vacuum appliance that can be used as a vacuum mop, such that the suction air stream may contain moisture.

[0008] The problem is solved by the subject matter of the independent patent claim. Advantageous embodiments are in particular defined in the dependent patent claims, described in the following description or shown in the appended drawing.

[0009] With the foregoing and other objects in view there is provided, in accordance with the invention, a suction unit for a handheld vacuum appliance. The suction unit contains a suction mouth, a separating unit, and a blower configured to generate a main suction air stream from the suction mouth via the separating unit to the blower of the suction unit. The suction unit further has a handle with a housing wall defining a cavity, and in the handle an electronic component is disposed. The cavity has at least one inlet opening formed in the housing wall. A bypass channel has an inlet facing toward the cavity and an outlet facing toward the blower. The blower is configured to generate a bypass air stream from the at least one inlet opening, through the cavity

and through the bypass channel to the blower. The cavity is covered, on a side facing toward the inlet of the bypass channel, by a shielding element configured to shield the cavity against moisture from the main suction air stream and/or from exhaust air from the blower. The shielding element has a sealed opening and in the sealed opening the inlet of the bypass channel is disposed.

[0010] In one aspect, a suction unit for a handheld vacuum appliance is described. The suction unit contains a blower which is configured to generate a main suction air stream from the suction mouth (or the suction tube) of the suction unit via the separating unit of the suction unit to the blower of the suction unit. If appropriate, a wet cleaning nozzle may be connected to the suction mouth or to the suction tube, such that the main suction air stream may include moisture (in addition to dust). The blower may be arranged above the separating unit as viewed along a vertical axis of the suction unit. The separating unit typically contains a collecting container for receiving suctioned dirt.

[0011] The suction unit furthermore contains a handle having a cavity which is formed by a housing wall and in which an electronics component is arranged (wherein the electronics component may be designed to control the blower). The cavity has one or more inlet openings in the housing wall. The handle may be designed to be gripped by a user using one hand for the purposes of holding the suction unit. The handle may be arranged behind the blower and the separating unit as viewed along a longitudinal axis of the suction unit (which extends perpendicularly to the vertical axis). The handle may (like the handle of a jug) extend along the vertical axis from the top side of the suction unit (at which the blower is arranged) to the bottom side of the suction unit (at which the energy store of the suction unit and/or the bottom side of the separating unit is arranged). The cavity having the electronics component may be arranged at the bottom side of the suction unit (directly above the holder for the energy store as viewed along the vertical axis).

[0012] The handle may be part of a housing of the suction unit. The blower may also be arranged in the housing. The housing may furthermore have a connection region which faces toward the side wall of the separating unit (with the collecting container) and which extends from the bottom side to the top side of the suction unit. The connection region may thus extend substantially along the vertical axis. The handle may, together with the connection region, form a ring. The connection region may have a coupling at which the separating unit can be fastened to the housing of the suction unit.

[0013] The suction unit furthermore contains a bypass channel which has an inlet facing toward the cavity and an outlet facing toward the blower. The bypass channel may extend within the connection region of the housing of the suction unit. Here, the bypass channel may extend along the vertical axis from the bottom side of the suction unit (at which the cavity is arranged) to the top side of the suction unit (at which the blower is arranged). The bypass channel may contain a bypass hose. In particular, the bypass channel may be formed in an efficient manner as a bypass hose.

[0014] The blower is configured to generate a bypass air stream from the one or more inlet openings, through the cavity and through the bypass channel to the blower. The blower is thus made capable of generating a bypass air stream in addition to and separately from the main suction air stream in order to achieve efficient and reliable cooling of the electronics component. Here, the bypass air stream has fresh ambient air, such that particularly reliable cooling can be achieved. Furthermore, the bypass air stream does not contain moisture from the main suction air stream, thus making careful cooling of the electronics component possible.

[0015] The electronics component may have a first edge facing toward the inlet of the bypass channel. The one or more inlet openings are preferably arranged at the second edge of the electronics component, the second edge facing away from the inlet of the bypass channel, wherein the first and the second edge of the electronics component are arranged opposite one another. Particularly reliable cooling of the electronics component can thus be achieved.

[0016] The electronics component has (in relation to the vertical axis) a first side (in particular a

bottom side) and a second side (in particular a top side). The one or more inlet openings may face toward the first side (in particular the bottom side) of the electronics component, and the inlet of the bypass channel may face toward the second side (in particular the top side) of the electronics component. Particularly reliable cooling of the electronics component by means of the bypass stream can thus be achieved.

[0017] The inlet of the bypass channel may constitute a constriction in relation to the cavity, such that the bypass air stream within the cavity is accelerated and/or compressed toward the inlet of the bypass channel. The cooling effect can thus be further improved.

[0018] The cavity is preferably covered, on the side facing toward the inlet of the bypass channel, by a shielding element which is configured to shield the cavity against moisture from the blower and/or from the main suction air stream and/or from exhaust air from the blower. Here, the shielding element preferably has a sealed opening (with an annular seal) in which the inlet of the bypass channel is arranged. Through the provision of a wall-like shielding element, the electronics component can be particularly reliably and efficiently protected against moisture.

[0019] The electronics component may have a circuit board (on which one or more electronic devices are arranged). The circuit board and the shielding element may be arranged at an acute angle with respect to one another, in particular at an angle of 75° or less. It is thus possible for the bypass air stream to be focused in order to achieve improved cooling of the electronics component.

[0020] As already stated, the blower may be arranged above the separating unit as viewed along the vertical axis of the suction unit. Furthermore, the cavity may be arranged below the blower as viewed along the vertical axis. The bypass channel may extend (within the connection region of the housing of the suction unit) along the vertical axis from the cavity to the inlet of the blower.

[0021] At least one portion of the bypass channel may extend above the inlet of the blower as viewed along the vertical axis. The portion of the bypass channel may be configured such that a flow of moisture through the portion of the bypass channel from the inlet of the blower to the inlet of the bypass channel is impeded. The electronics component can thus be particularly reliably protected against moisture.

[0022] The suction unit may be configured such that the exhaust air from the blower is conducted (within the connection region of the housing) along the vertical axis, along the bypass channel and along the side wall of the separating unit to an outlet opening that is arranged below the blower as viewed along the vertical axis, in particular on the bottom side of the suction unit. Here, by means of the shielding element, it can be achieved that no moisture from the exhaust air enters the cavity having the electronics component. The convenience of the suction unit can be increased by virtue of the exhaust air flowing out at the bottom side of the suction unit.

[0023] The suction unit may thus be designed such that the exhaust air from the blower and the bypass air stream run adjacent to one another within the connection region of the housing of the suction unit. The exhaust air and the bypass air stream may in this case have opposite flow directions. In particular, the exhaust air may flow along the vertical axis from the top side toward the bottom side of the suction unit. On the other hand, the bypass air stream may flow along the vertical axis from the bottom side toward the top side of the suction unit. The bypass air stream is conducted within the bypass channel and is therefore separate from the exhaust air (which, for example, flows freely to the bottom side of the suction unit within the connection region of the housing). Particularly efficient and reliable air guidance within the housing of the suction unit can thus be achieved.

[0024] The blower may be embedded in a blower capsule, wherein the blower capsule has a connection element. The outlet of the bypass channel may be mechanically connected to the connection element, in particular fitted onto the connection element. Efficient and reliable fixing of the bypass channel can thus be achieved.

[0025] As already stated, the suction unit may have an electrical energy store which is arranged at the one or more inlet openings such that the bypass air stream flows past at least a subregion of the

energy store. Particularly efficient cooling of the electrical energy store of the suction unit can thus be achieved.

[0026] The suction mouth of the suction unit has a main cross-sectional area, and the bypass channel has a bypass cross-sectional area. The main cross-sectional area is preferably larger than the bypass cross-sectional area by a factor of 10 or more, in particular by a factor of 100 or more. As a result, the bypass channel does not result in any significant impairment of the suction power of the suction unit.

[0027] The housing wall that forms the cavity may have, on the inner side facing toward the cavity, one or more guide fins and/or guide protuberances that are each designed to conduct the bypass air stream through the cavity past the electronics component. Alternatively or in addition, the circuit board may comprise one or more drilled holes and/or cutouts through which a proportion of the bypass air stream flows when the blower is in operation. The cooling of the electronics component can thus be further improved.

[0028] According to a further aspect, a suction unit for a handheld vacuum appliance is described, wherein the suction unit comprises a blower which is configured to generate a main suction air stream from the suction mouth of the suction unit via a separating unit of the suction unit to the blower of the suction unit. It should be noted that the features described in this document are also applicable individually or in combination to this suction unit.

[0029] The suction unit contains a housing having a housing wall which forms a cavity in which an electronics component is arranged, the electronics component being designed to control the blower, for example. The cavity has one or more inlet openings in the housing wall.

[0030] The suction unit furthermore contains a bypass channel which has an inlet facing toward the cavity and an outlet facing toward the blower. The blower is configured to generate a bypass air stream from the one or more inlet openings, through the cavity and through the bypass channel to the blower.

[0031] The inlet of the bypass channel preferably constitutes a constriction in relation to the cavity, such that the bypass air stream within the cavity is accelerated and/or compressed toward the inlet of the bypass channel. Particularly reliable and efficient cooling of the electronics component of the suction unit can thus be achieved.

[0032] According to a further aspect, a vacuum appliance, in particular a handheld vacuum cleaner (with mop function), is described, which comprises the suction unit described in this document.

[0033] It should be noted that all aspects of the suction unit described in this document and of the vacuum appliance described in this document may be combined with one another in a variety of ways. In particular, the features of the patent claims may be combined with one another in a variety of ways.

[0034] Other features which are considered as characteristic for the invention are set forth in the appended claims.

[0035] Although the invention is illustrated and described herein as embodied in a suction unit with cooling using fresh air, it is nevertheless not intended to be limited to the details shown, since various modifications and structural changes may be made therein without departing from the spirit of the invention and within the scope and range of equivalents of the claims.

[0036] The construction and method of operation of the invention, however, together with additional objects and advantages thereof will be best understood from the following description of specific embodiments when read in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0037] FIG. 1 is a diagrammatic, perspective view of an exemplary vacuum appliance having a

suction unit, a suction tube and a nozzle;

[0038] FIG. 2A is a section view of a suction unit having a bypass channel;

[0039] FIG. 2B is a section view of an electronics component of the suction unit;

[0040] FIG. 2C is a sectional view of a main suction air stream and a bypass air stream of the suction unit;

[0041] FIGS. 3A and 3B are different perspective views of the blower capsule; and

[0042] FIG. 4 is a perspective view of an exemplary shielding element for shielding the electronics component.

DETAILED DESCRIPTION OF THE INVENTION

[0043] As discussed in the introduction, the present document is concerned with achieving efficient and reliable cooling of the electronics of a suction unit of a vacuum appliance that can also be used as a vacuum mop. In this context, FIG. 1 shows an exemplary (handheld) vacuum cleaner **100** (as an example of a vacuum appliance) containing a suction unit **110** that has an electrical energy store **111**. The suction unit **110** has a handle **112** that a user can grip using one hand in order to hold the suction unit **110**. The blower of the suction unit **110** generates a suction air stream through the suction mouth **114** of the suction unit **110** via a separating unit **113** of the suction unit **110** to the blower. The suction unit **110** may be configured to be used on its own as a vacuum appliance.

[0044] An accessory part **120**, **130** can be connected by means of a coupling **121** to the suction unit **110**. In the example illustrated, the suction unit **110** is connected by means of a coupling **121** to a suction tube **120**, which in turn is connected by means of a coupling **121** to a floor nozzle **130**. The floor nozzle **130** may be designed as a wet cleaning nozzle.

[0045] The suction air stream generated by the blower of the suction unit **110** may be used to cool an electronics component and/or the energy store **111** of the suction unit **110**. If a wet cleaning nozzle is used, the suction air stream may contain moisture, which can damage the electronics component and/or the energy store **111**.

[0046] FIGS. 2A to 2C show different views of a suction unit **110** which has a bypass channel **210** fluidically coupled to the blower **200**, such that the blower **200** can generate a bypass air stream **220** in the bypass channel **210**. The bypass channel **210** may be formed by a hose. The bypass channel **210** extends from the blower **200** through the housing of the suction unit **110** in the direction of the electronics component **221** and/or the energy store **111** of the suction unit **110**, such that the bypass air stream **220** flows past the electronics component **221** and/or past the energy store **111**.

[0047] As can be seen in particular from FIG. 2B, the electronics component **221** may be shielded from the blower **200** by means of a shielding element **230**, in particular in order to prevent moisture from passing from the blower **200** and/or from the exhaust air **241** from the blower **200** to the electronics component **221**. The bypass channel **210** may be arranged on that side of the shielding element **230** which faces toward the blower **200**, and the electronics component **221** may be arranged on that side of the shielding element **230** which faces away from the blower **200**. The shielding element **230** may have a sealed opening **231** in which the inlet **211** of the bypass channel **210** is arranged. It can thus be achieved that the bypass air stream **220** passes via the electronics component **221**, through the opening **231** of the shielding unit **230** and into the inlet **211** of the bypass channel **210**.

[0048] The electronics component **221** is enclosed by the housing wall of the handle **112** of the suction unit **110**. The housing wall has one or more entry openings **213** via which the bypass air stream **220** enters the housing from the outside and arrives at the electronics component **221**. The one or more entry openings **213** are arranged at that side of the electronics component **221** which faces away from the inlet **211** of the bypass channel **210**, such that the bypass air stream **220** flows from the one or more entry openings **213** via the electronics component **221** to the inlet **211** of the bypass channel **210**, and in so doing cools the electronics component **221**.

[0049] As can be seen from FIG. 2C, the blower **200** thus generates not only the main suction air

stream **240** (which flows through the separating unit **113** of the suction unit **110**, as illustrated in FIG. 2A) but also the bypass air stream **220** through the bypass channel **210**.

[0050] FIGS. 3A and 3B show different views of the blower capsule **300** which encloses the blower **200**. The blower capsule **300** may have a connection element **310** for the connection of the outlet **212** of the bypass channel **210**. The outlet **212** of the bypass channel **210** can for example be pushed onto the connection element **310** in order to fix the outlet **212** of the bypass channel **210** to the blower capsule **300**. The bypass air stream **220** can thus be generated in a particularly stable manner.

[0051] FIG. 4 shows an exemplary shielding element **230** having the sealed opening **231** for the inlet **211** of the bypass channel **210**.

[0052] It is thus possible for the electronics **221** in the handle **112** of the suction unit **110** to be cooled by means of a supply of fresh air that is drawn in through a secondary air channel (that is to say through the bypass channel **210**). The collecting container (as part of the separating unit **113**), the blower **200**, the handle **112**, the control electronics **221** for the blower **200** and/or a holder for the electrical energy store **111** may be integrated in the suction unit **110** (that is to say in the handheld part of the vacuum appliance **100**). The air stream **240** that is supplied from the tube (that is to say the suction mouth **114**) is conducted through the collecting container and the filter of the separating unit **113** to the blower **200**. The exhaust air **241** from the blower **200** may furthermore be conducted onward into the handle **112** and/or through the connection region of the housing of the suction unit (to the bottom side of the suction unit **110**; as illustrated by way of example in FIG. 2A).

[0053] The cooling of the electronics **221** is effected by means of a bypass channel **220** through which fresh air from the outside is drawn into the handle **112**. A bypass channel **210** (in particular a bypass hose) is positioned close to the inlet of the blower **200**. During operation, the negative pressure that is generated by the blower **200** at the inlet of the blower draws in an air stream **220** from the bypass channel **210**. The air that is drawn into the bypass channel **210** at the channel inlet side **211** originates from a (largely) closed-off cavity of the handle **112**, in which the electronics **221** are arranged. Fresh air flows in through one or more entry openings **213** on the housing wall close to the holder for the energy store **111** and flows around the electronics **221**. That region of the handle **112** in which the electronics **221** are positioned may be separated from the rest of the interior of the handle **112** by partitions (that is to say by a shielding element **230**) such that the exhaust air **241** from the blower **200** which is conducted by the blower **200** into the handle **112** and/or into the connection region of the housing of the suction unit **110** does not reach the electronics **221**, and therefore cannot mix with the fresh air, before exiting the suction unit **110** via exit openings in the housing of the suction unit **110** (on the bottom side of the suction unit **110**).

[0054] By virtue of the one or more entry openings **213** for the fresh air being arranged in the region of the holder for the energy store **111**, simultaneous cooling of the electronics **221** and of the energy store **111** is achieved. Here, air which is at ambient temperature and which is therefore significantly cooler than the exhaust air **241** from the blower **200** flows both around the housing of the energy store **111** and around the electronics **221**. In the cavity of the handle **112** having the integrated electronics **221**, the fresh air flowing in to the bypass channel **210** is conducted around the electronics **221** and, in the process, flows over large parts of the board (that is to say the circuit board) of the electronics **221**, or over large parts of the electronics **221**, rather than being directed only locally onto a small portion. The one or more entry openings **213** for the fresh air are preferably situated laterally in the housing below the electronics board, whereas the inlet side **211** of the bypass channel **210** is situated above the electronics board. The air that is drawn in to the bypass channel **210** thus flows around the entire board before being focused on the inlet **211** of the bypass channel **210**. The electronics components arranged in this portion of the electronics **221** are thus cooled particularly effectively, because the flow velocity and the flow density in the vicinity of the inlet **211** of the bypass channel **210** are increased.

[0055] The partition **230** (that is to say the shielding element) situated in the hand **112** at the inlet **211** of the bypass channel **210** is preferably arranged at a certain angle (obliquely) with respect to the electronics board **221**, such that the board **221** and the partition **230** enclose an acute angle. This arrangement reduces the volume of the cavity in the unused region and assists the supply of air into the bypass channel **210** and the concentration of the air in the region close to the inlet **211** of the bypass channel **210**.

[0056] The flow rate of the fresh air that is drawn in for the purposes of cooling the electronics **221** can be set by means of the internal diameter of the bypass channel **210**. The ratio of the internal diameter of the bypass channel **210** to the internal diameter of the suction tube **114** (that is to say of the suction mouth) substantially defines the flow rate of the fresh air. The internal diameter of the bypass channel **210** may for example be 3 mm, and the internal diameter of the suction tube **114** may be 30 mm, such that the area of the cross section of the bypass channel **210** is for example one hundredth of the area of the cross section of the suction tube **114**.

[0057] The air stream **220** from the bypass channel **210** exits the bypass channel **210** at a coupling means **310** to the side of the blower **200**. The air **220** is conducted from the coupling means to the inlet of the blower **200**, where said air mixes with the air **240** drawn in from the collecting container and is drawn into the blower **200**. Owing to the elevated position of the outlet **212** of the bypass channel **210** relative to the inlet of the blower **200** and the associated increased distance of the bypass channel **210** from the moisture-carrying air stream **240** from the collecting box, no additional measures such as filters are necessary to protect the bypass channel **210** against moisture or dust.

[0058] A dimensionally stable material (such as PE (polyethylene)) may be used for the bypass channel **210**, in particular for the bypass hose. Fixing ribs which additionally stabilize the bypass channel **210** in terms of its shape are preferably incorporated into the housing shells of the handle **112**. By means of the fixing ribs, the bypass channel **210** is routed, and its ends **211**, **212** are preferably oriented, such that the bypass channel **210** is inclined in the flow direction, and therefore no significant noise generation occurs and no significant flow resistance is generated.

[0059] The cross section of the bypass channel **210** may be selected (for example with an external diameter of 5 mm and an internal diameter of 3 mm) so as to ensure an optimum throughput of air for the purposes of cooling the electronics, and at the same time minimize possible vibrations of the bypass channel **210**. Furthermore, the wall thickness (for example of 1 mm) of the bypass channel **210** ensures dimensional stability and resistance to distortion with respect to the external heating caused by the heated exhaust air **241** which flows along the outside of the bypass channel **210**.

[0060] That end **211** of the bypass channel **210** which is oriented toward the electronics **221** is preferably (for the purposes of protecting against moisture) fixed directly in the rubber seal of the electronics shield **230**. That end **212** of the bypass channel **210** which leads to the blower **200** may be connected by a flexible elastic coupling means **310** (for example an intermediate piece in the form of a silicone hose, for example) to the blower capsule **300**. The bypass channel **210** is thus decoupled from vibrating parts (for example the blower **200**) and can absorb vibrations (for example if the appliance is dropped) without any impairment of function.

[0061] The cooling air stream **220** is free from drawn-in moisture and may optionally also cool the housing of the energy store **111**, because fresh air is drawn in from the outside. Even during operation with a damaged or inadvertently uninstalled filter of the separating unit **113**, the electronics **221** cannot come into contact with moisture from the nozzle **130** and/or with suctioned dust, because the electronics are separated from the exhaust air **241** by partitions **230** (that is to say by the shielding element).

[0062] Additional fins or protuberances may be integrated in the cavity of the electronics **221** and provide enhanced or improved guidance of air around the electronics **221**. The air stream **220** within the cavity can thus be directed to parts of the electronics **221** requiring particularly intense cooling and can be concentrated and accelerated there, or at another location may be conducted

more widely over large areas of the electronics board.

[0063] The electronics board itself may be provided with drilled holes or cutouts in order to improve the air flow in the region of the electronics and/or in order to conduct air to desired locations that are also situated within the extent of the board but on the other side of the board.

[0064] The measures described in this document have the effect that the electronics **211** are cooled by means of fresh ambient air from the outside, which is typically cooler than the exhaust air **241** from the blower **200**. It is possible to implement simultaneous cooling of the energy store **111** and of the electronics **221**, because the air **220** that is drawn into the housing through the inlet openings **213** also flows over the housing of the energy store **111**. Cooling of the entire electronics board can be achieved because the fresh air flows around the entire board and does not merely (locally) draw away warm air.

[0065] The bypass air stream **220** for cooling purposes is separate from the main suction air stream, such that moisture and/or dust/dirt cannot enter the bypass air stream **220**. The electronics **221** are thus protected against moisture and are not situated in the space that is exposed to dust. The electronics **221** cannot come into contact with dust even if a filter of the separating unit **113** is damaged or uninstalled. The fresh air stream **220** furthermore ensures continuous suctioning of dust away from the electronics **221**.

[0066] The heated air dissipated from the electronics **221** is supplied via the bypass channel **210** to the blower **200** and is mixed, in the intake region (that is to say at the inlet) of the blower **200**, with the air **240** that has been drawn in via the suction channel. From the blower **200** onward, the air stream is therefore warmer than the rest of the ambient air. Since heated air can absorb more moisture than relatively cool air, the moisture contained in the intake air can be more effectively retained in the air stream. The tendency for the moisture to condense within the suction unit **110** is thus reduced, such that an impairment of the suction unit **110**, in particular of the blower **200**, by moisture can be reduced.

[0067] What is described, therefore, is cooling of an electronics board **221** in a handheld vacuum cleaner **100**, **110**, in which the power supply and the electronics are arranged on/in the handle **112**, by means of a bypass channel **210**. The one or more cooling air openings **213** are also situated in the handle **112**.

[0068] Fresh air **220** that is drawn in from the outside flows through the bypass channel **210** in order to cool the electronics **221** and the energy store **111**, and therefore an additional structural unit for separating moisture out of the main suction air stream **240**, and/or a coating of the electronics **221**, can be avoided.

[0069] When the blower **200** is in operation, cooling air **220** is drawn in through the openings **213** on the handle **112** and is conducted via the energy store **111** and the electronics **221** in order to cool these. Here, the cooling air **220** can flow freely over the electronics **221**. Via a (sealed) constriction **231** arranged downstream (downstream of the electronics **221**), the heated air is supplied via a bypass channel **210** (in particular a hose) to the blower **200**. The supply takes place upstream of the blower **200** together with the main suction air stream **240**. The cooling air is mixed with the dust-laden air (that is to say the main suction air stream **240**) in the blower **200** and is then released to the external surroundings via the openings in the handle **112** or in the connection region of the housing of the suction unit **110**. The constriction **231** arranged downstream allows cooling air to flow around the entirety of the electronics **221**. The bypass channel **210** allows the cooling air to be conducted with low resistance.

[0070] What is described, therefore, is a handheld and/or battery-powered vacuum cleaner **100** that has a motor-blower unit **200** for generating suction air **220**, **240**. The vacuum cleaner **100** contains a main air channel from an entry opening (for example a nozzle **130**) via a dust-separating unit **113** to the motor-blower unit **200**. The vacuum cleaner **110** furthermore contains a secondary air channel **210** from an entry opening **213** that is arranged on the housing of the vacuum cleaner **100**. The vacuum cleaner **100** furthermore has electronics **221**, which are to be cooled, in the housing,

wherein the electronics **221** are arranged in the secondary air channel.

[0071] A constriction **231** may be arranged in the secondary air channel **210**, wherein the constriction **231** is arranged downstream of the electronics **221** that are to be cooled. The cooling air **220** can be supplied directly from the constriction **231** to the blower **200**. Here, the cooling air may be discharged via a hose. The constriction may be formed by the opening or the inlet **211** of the hose.

[0072] The opening **213** of the cavity may be arranged on the handle **112** of the vacuum cleaner **100**. The secondary air **220** may be drawn in above the energy store **111** (through the opening **213**), and may be such that the secondary air **220** cools the energy store **111**.

[0073] The entry opening **310** for the cooling air at the end **212** of the channel **210** may be arranged on the blower capsule **300** of the motor-blower unit **200** such that the cooling air **220** is mixed with the main suction air stream **240** directly upstream of a safety guard element of the blower capsule **300**.

[0074] The present invention is not restricted to the exemplary embodiments shown. In particular, it should be noted that the description and the figures are intended merely to illustrate the principle of the suction unit **110** and/or of the vacuum appliance **100**.

[0075] The following is a summary list of reference numerals and the corresponding structure used in the above description of the invention: [0076] **100** Vacuum appliance (vacuum mop) [0077] **110** Suction unit [0078] **111** Electrical energy store [0079] **112** Handle [0080] **113** Separating unit [0081] **114** Suction mouth [0082] **120** Accessory part (suction tube) [0083] **121** Coupling [0084] **130** Nozzle (e.g. wet cleaning nozzle) [0085] **200** Blower [0086] **210** Bypass channel [0087] **211** Inlet (bypass channel) [0088] **212** Outlet (bypass channel) [0089] **213** Entry opening [0090] **220** Bypass air stream [0091] **221** Electronics component [0092] **230** Shielding element [0093] **231** Opening [0094] **240** Main suction air stream [0095] **241** Exhaust air [0096] **300** Blower capsule [0097] **310** Connection element (for bypass channel)

Claims

1. A suction unit for a handheld vacuum appliance, the suction unit comprising: a suction mouth; a separating unit; a blower configured to generate a main suction air stream from said suction mouth via said separating unit to said blower of the suction unit; a handle having a housing wall defining a cavity, and in said handle an electronic component is disposed, wherein said cavity has at least one inlet opening formed in said housing wall; a bypass channel having an inlet facing toward said cavity and an outlet facing toward said blower; said blower configured to generate a bypass air stream from said at least one inlet opening, through said cavity and through said bypass channel to said blower; and a shielding element, wherein said cavity is covered, on a side facing toward said inlet of said bypass channel, by said shielding element configured to shield said cavity against moisture from the main suction air stream and/or from exhaust air from said blower, and wherein said shielding element has a sealed opening formed therein and in said sealed opening said inlet of said bypass channel is disposed.

2. The suction unit according to claim 1, wherein: said electronic component has a first edge facing toward said inlet of said bypass channel; said at least one inlet opening is disposed at a second edge of said electronic component, said second edge facing away from said inlet of said bypass channel; and said first edge and said second edge of said electronic component are disposed opposite one another.

3. The suction unit according to claim 1, wherein: said electronic component has a first side and a second side situated opposite said first side; said at least one inlet opening faces toward said first side of said electronic component; and said inlet of said bypass channel faces toward said second side of said electronic component.

4. The suction unit according to claim 1, wherein said inlet of said bypass channel constitutes a

constriction in relation to said cavity, such that the bypass air stream within said cavity is accelerated and/or compressed toward said inlet of said bypass channel.

5. The suction unit according to claim 1, wherein: said electronic component has a circuit board; and said circuit board and said shielding element are disposed at an acute angle with respect to one another.

6. The suction unit according to claim 1, wherein: said blower is disposed above said separating unit as viewed along a vertical axis of the suction unit; said cavity is disposed below said blower as viewed along the vertical axis; and said bypass channel extends along the vertical axis from said cavity to an inlet of said blower.

7. The suction unit according to claim 6, wherein at least one portion of said bypass channel extends above said inlet of said blower as viewed along the vertical axis, such that a flow of moisture through said portion of said bypass channel from said inlet of said blower to said inlet of said bypass channel is impeded.

8. The suction unit according to claim 6, wherein the suction unit is configured such that the exhaust air from said blower is conducted along the vertical axis, along said bypass channel and along a side wall of said separating unit to an outlet opening that is disposed below said blower as viewed along the vertical axis.

9. The suction unit according to claim 1, wherein said bypass channel has a bypass hose.

10. The suction unit according to claim 1, further comprising a blower capsule, said blower is embedded in said blower capsule, said blower capsule has a connection element, and said outlet of said bypass channel is mechanically connected to said connection element.

11. The suction unit according to claim 1, wherein the suction unit has an electrical energy store which is disposed adjacent to said at least one inlet opening such that the bypass air stream flows past at least a subregion of said electrical energy store.

12. The suction unit according to claim 1, wherein: said suction mouth has a main cross-sectional area; said bypass channel has a bypass cross-sectional area; and said main cross-sectional area is larger than said bypass cross-sectional area by a factor of at least 10.

13. The suction unit according to claim 1, wherein said housing wall that forms said cavity has, on an inner side facing toward said cavity, at least one guide fin and/or guide protuberance that are each configured to conduct the bypass air stream through said cavity past said electronic component.

14. The suction unit according to claim 1, wherein: said electronic component has a circuit board; and said circuit board has at least one drilled hole and/or cutout formed therein and through which a proportion of the bypass air stream flows when said blower is in operation.

15. The suction unit according to claim 5, wherein the acute angle is an angle of 75° or less.

16. The suction unit according to claim 10, wherein said outlet of said bypass channel is mechanically fitted on said connection element.

17. The suction unit according to claim 12, wherein said main cross-sectional area is larger than said bypass cross-sectional area by a factor of at least 100.
